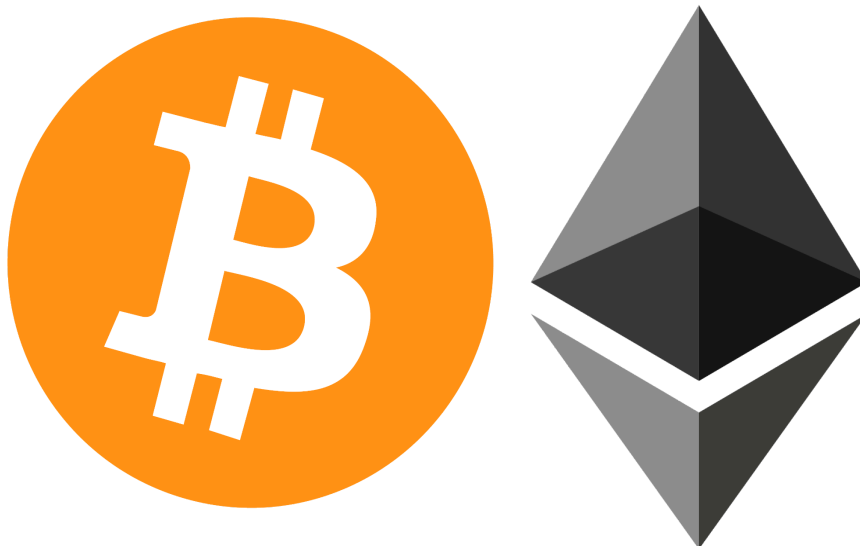




Project Plan & Proposal

Multivariate Forecasting and Predictability Analysis of Cryptocurrency
Market Dynamics

2026 Capstone Group 27 A

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1.0 Company & Project Context

1.1 About the Company

The Data Science Institute is the client organisation for this project. It focuses on applying data science, machine learning, and advanced analytics techniques to better understand complex datasets and generate meaningful insights. The institute works with researchers, students, and industry partners to explore how data-driven approaches can be used to address real-world analytical challenges.

In this project, the Data Science Institute has engaged our team to analyse cryptocurrency market data and investigate the predictability of key market variables such as price, returns, volatility, and trading volume using data mining and machine learning techniques.

1.2 Mission Statement

Our mission is to deliver reliable and insightful data analytics and predictive modelling solutions that provide businesses and organisations with a better understanding of complex data, allowing them to make more informed and appropriate decisions. By applying techniques such as data analysis, machine learning, and data mining, we intend to transform raw data into meaningful insights that support improved performance, innovation, and strategic growth.

Through this project, our goal is to support the Data Science Institute by **analysing the predictability of cryptocurrency market behaviour** using data mining models applied to key market variables including price, returns, volatility, and trading volume.

2.0 Our Team & Roles

2.1 The Team & Roles

Carlton Hogan - Administrative Team Leader

Carlton is a Business and Information Technology student at the University of Technology Sydney, majoring in Finance with a focus on data analytics. His background combines financial analysis with statistical modelling and machine learning applications in finance. In this project, he will contribute to model development, analytical design, and overall project coordination.

Carlton is responsible for coordinating the administrative side of the project and supporting the team's overall progress. His role includes organising meetings, preparing agendas, recording minutes, and monitoring progress against the timeline. He will also contribute to model development and assist with writing the final report.

Sambhav Gurung - Experimental Analysis

Sambhav is a Computer Science Honours student at the University of Technology Sydney, majoring in Artificial Intelligence. His background includes software engineering, machine learning, and computational problem solving. He has experience with programming languages such as Java, Python, and C++, as well as machine learning tools including KNIME.

Sambhav is responsible for designing and conducting experiments to evaluate the forecasting models. He will prepare training and testing data, run model evaluations, and analyse performance metrics. He will also help interpret the results to determine which models produce the most reliable predictions.

Kevin Nochez - Model Developer

Kevin is an Information Technology student at the University of Technology Sydney, majoring in Data Analytics. His academic background includes data analysis, statistical modelling, machine learning, and artificial intelligence. He has experience with Python, C, and Java, and is particularly interested in data analysis and machine learning applications.

Kevin is responsible for developing and implementing the forecasting models used in the project. He will select suitable modelling approaches and build models to predict variables such as price, return, volatility, and trading volume. He will also evaluate and compare model performance to identify which variables can be predicted most effectively.

Huijoon Jang - Head of Reporting

Huijoon is an Information Technology student at the University of Technology Sydney, majoring in Data Analytics. His background includes data processing, statistical analysis, and identifying patterns in complex datasets. He has experience with Python and SQL and is particularly interested in data visualisation and predictive modelling.

Huijoon is responsible for overseeing the project's written documentation and ensuring the final report is clear, consistent, and professionally structured. He will coordinate report sections from team members, review content quality, and help present results in a logical and understandable way. He will also assist in summarising key findings from the analysis.

Alexandros Mullane - Machine Learning Development & Reporting

Alexandros is studying a Bachelor of Artificial Intelligence at the University of Technology Sydney. Through his studies, he has developed skills in programming, data handling, and analytical thinking, with a growing focus on machine learning and artificial intelligence. In this project, he will contribute to research, data analysis, and solution development.

Alexandros is responsible for developing machine learning models for multivariate time series forecasting, with a focus on LSTM and GRU architectures. He will help evaluate predictive performance across variables such as price, return, volatility, and trading volume.

He will also contribute to the report by explaining the methodology, presenting results, and discussing findings.

Alvin Ji Shan Tiong - Data Collector and Analyst

Alvin is an Information Technology student at the University of Technology Sydney, majoring in Data Analytics. He has experience in data analysis, statistical methods, and Python-based data processing, with an interest in machine learning and time-series analysis. In this project, he will contribute to data collection, preprocessing, and forecasting model implementation.

Alvin is responsible for collecting, organising, and preparing the cryptocurrency datasets used in the project. He will support data preprocessing, feature engineering, and exploratory data analysis to help the team understand patterns in the data. He will also contribute to the implementation and evaluation of forecasting models.

3.0 The Project

3.1 Project Context

This project investigates the extent to which data mining models can predict key time-series variables in a non-stationary environment, using the cryptocurrency market as a testbed. Cryptocurrency markets are well suited to this purpose because they are characterised by high volatility, rapid regime shifts, and substantial uncertainty. These features create a challenging setting for evaluating the capabilities and limitations of predictive models.

Our Project is centred around forecasting 4 key market variables:

1. Price
2. Return
3. Volatility
4. Trading Volume

These variables capture important aspects of market behaviour and allow the study to assess predictive performance across different dimensions of the cryptocurrency market.

To address this problem, the project will evaluate regression models, random forest, and neural networks, with an LLM-based approach considered as an optional extension if feasible. The final outputs will include a written report, GitHub code repository, and presentation.

3.2 How we can help

As part of the Data Science Institute, our team aims to apply data mining and predictive modelling techniques to explore the predictability and volatility of cryptocurrency market behaviour. By organising and analysing historical market data from cryptocurrencies such as

Bitcoin and Ethereum, we seek to identify patterns and correlations between pivotal market variables like price, return, volatility and trading volume.

Using technical approaches ranging from statistical analysis, machine learning models, to neural networks, we will evaluate how accurately these market variables can be predicted. By successfully predicting these variables, we aim to help understand and showcase which aspects of the market show more consistent patterns and predictability and which are more difficult to predict due to market volatility and uncertainty.

3.3 The Data Mining Problem

The business problem addressed in this project is understanding the behaviour and predictability of cryptocurrency markets. Cryptocurrency markets are well known for their extremely high volatility, rapid regime changes and strong uncertainty. Due to these characteristics, predicting cryptocurrency market behaviour is considered more difficult than predicting traditional financial market behaviour.

Unlike traditional financial datasets where observations are often independent, cryptocurrency market data forms a complex time-series dataset with strong temporal dependencies. Market variables such as price, return, volatility, and trading volume are influenced by past market behaviour and can exhibit nonlinear dependencies. In addition, cryptocurrency markets often display non-stationary behaviour, and this makes statistical properties such as mean and variance change over time.

Thus, from a data mining perspective we can formulate the problem as a multivariate time-series forecasting task. The objective is to use historical market data to predict future values of several key market variables. For this project, we will forecast four cryptocurrency market variables: price, return, volatility, and trading volume.

We will explore and compare a range of different modelling approaches for this problem that include statistical time-series models, machine learning algorithms, and deep learning architectures. By evaluating forecasting performance across models and variables, this project aims to analyse which aspects of cryptocurrency market behaviour are more predictable, and which remain largely dominated by uncertainty.

4.0 Project Proposal

4.1 Project Plan

This project follows a clear workflow to test how well data mining models can predict cryptocurrency market behaviour. The analysis focuses on four target variables: **price, return, volatility, and trading volume**. Using historical data from Bitcoin and Ethereum, the project will compare baseline, machine learning, and neural network models to assess which variables are more predictable.

Data collection

The dataset used in this project was obtained from Binance's public market data API. An application programming interface (API) is a set of rules or protocols that allows software applications to communicate and exchange data (Goodwin, 2024). This source was selected because it provides historical cryptocurrency trading data for our targeted cryptocurrencies, Bitcoin and Ethereum. The dataset covers the period from 2021-01-01 to 2025-12-31 and is recorded at daily intervals.

Historical market data was collected using a Python script that accessed the API to retrieve daily candlestick (kline) data for the BTCUSDT and ETHUSDT trading pairs. Specifically, the script automatically downloaded the data for the specified time range and extracted key trading variables from the API response. The raw data was then processed and organised into an XLSX dataset with separate sheets for each cryptocurrency. The final dataset contains 10 columns representing key market variables, including date, open, high, low, close, trading volume, quote asset volume, number of trades, taker buy base asset volume, and taker buy quote asset volume.

Data preprocessing and analysis

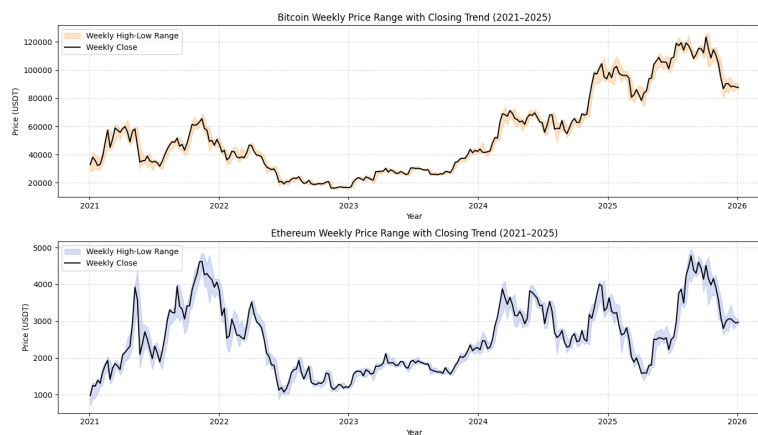


Figure 1 - Historical Price Range and Closing Trend for BTC and ETH

Before building the models, the dataset will undergo preprocessing to ensure the data is clean and consistent. The first step is to check for missing values, inconsistencies, and any potential anomalies in the historical trading data obtained from the Binance API.

Then, feature engineering will be performed to create extra variables from the original market data that may be useful for the prediction tasks of this project. Initially, these include daily and logarithmic returns, rolling volatility measures, moving averages (MA), exponential moving averages (EMA), momentum indicators, volume-based moving averages, and the Relative Strength Index (RSI) calculated over different time windows. Although these features are added to the dataset, not all of them may necessarily be useful during the modelling and experimentation stages.

In addition, exploratory data analysis (EDA) will be conducted to gain a better understanding of the historical behaviour of the cryptocurrency market. It is an important step in data analysis because it helps visualise data, identify patterns, and understand relationships between variables (GeeksforGeeks, 2021). Figure 1 provides an example of the visual analysis used in this project. It shows the historical price range and closing trends of Bitcoin and Ethereum between 2021 and 2025. Such visualisations help identify market trends and volatility patterns before model development. These patterns will provide useful insights for modelling the project's target variables: price, return, volatility, and trading volume.

Develop the models

In this stage, the main forecasting models will be developed. The project will compare a baseline regression model, a Random Forest model, and a neural network model such as **LSTM or GRU**.

This model set is appropriate because recent cryptocurrency forecasting research compares both ensemble-style machine learning methods and recurrent neural network models, while also showing that hybrid or more complex models do not always outperform simpler baselines in every case.

Conduct the experiments

After the models are developed, experiments will be run to test forecasting performance. The data will be split into training and testing periods so that each model can be evaluated on unseen observations.

Each model will be applied across the four forecasting targets: price, return, volatility, and trading volume. The purpose is to compare performance across model types and across variables, rather than assume that one model is best in every case. This follows the logic of recent comparative studies in cryptocurrency forecasting.

Analyze the experimental results

Once the experiments are complete, the results will be analysed to compare model performance. This stage will help identify which models perform best and which variables appear more predictable.

Performance will be measured using standard forecasting metrics such as **MAE** and **RMSE**, and where relevant the analysis may also consider whether a model captures the correct direction of movement. Prior research shows that model rankings can vary across cryptocurrencies and that lower forecast error does not always mean stronger practical prediction performance.

Discussion

The final stage will interpret the results and explain what they mean. It will discuss the effectiveness of each model and the challenges of forecasting cryptocurrency markets in a volatile and non-stationary environment.

This section will focus on whether some variables are more predictable than others, whether more complex models actually improve results, and what the findings suggest about the limits of prediction in cryptocurrency markets. This is consistent with the literature, which shows that forecasting performance depends on the asset, the period, and the structure of the data

4.2 Milestones

The project is structured around a series of internal milestones aligned with the Gantt chart, together with three formal assessment deadlines. In addition to these scheduled milestones, the team will bring a tangible deliverable to the Thursday meeting each week so that progress can be reviewed consistently and any issues can be identified early.

Internal milestones

- **March 2026** → Project commencement
- **1-15 March 2026** → Business understanding, scope confirmation, team role allocation, and supervisor consultation
- **8-18 March 2026** → Data collection, confirmation of the time horizon, and organisation of the raw dataset
- **18-29 March 2026** → Data cleaning, feature creation, and initial visual and descriptive analysis
- **26 March-15 April 2026** → Development of baseline, Random Forest, and LNN/LSTM models
- **6-27 April 2026** → Experiment setup, model testing, and refinement of outputs
- **20 April-3 May 2026** → Evaluation of results and comparison across target variables
- **1-17 May 2026** → Final report drafting, revision, and presentation preparation

External milestones

- **15 March 2026** → Assignment 1 Plan Proposal due
- **12 April 2026** → Assignment 2 Mid-Project Update due
- **17 May 2026** → Assignment 3 Final Report due

4.3 Resources Required

The project requires historical cryptocurrency market data from Binance, Python-based tools for data preprocessing and modelling, and access to standard machine learning libraries such as pandas, scikit-learn, and TensorFlow or PyTorch. It also requires GitHub for version control, shared documents for collaboration, and team laptops or university computing resources to run experiments and prepare the final report and presentation.

4.4 Gantt Chart

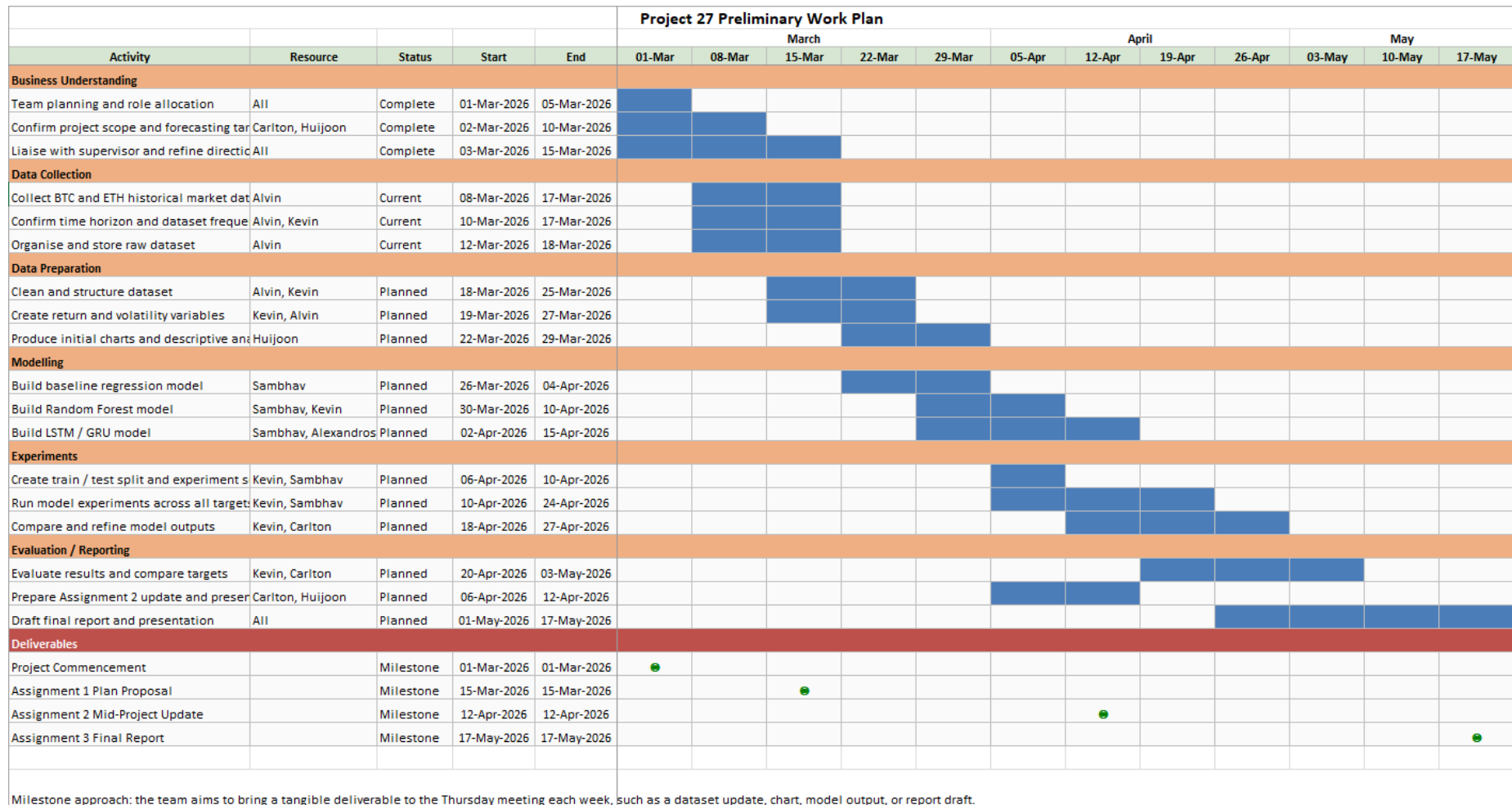


Figure 2 - Gantt Chart Showing the Current Work Plan (Note: Subject to Change)

5.0 References

- Binance n.d., Binance Data Collection, Binance Data Portal, viewed 15 March 2026, <<https://data.binance.vision/?prefix=data/spot/daily/klines/BTCUSDT/1h/>>.
- Bouteska, A, Abedin, MZ, Hajek, P & Yuan, K 2024, 'Cryptocurrency price forecasting – A comparative analysis of ensemble learning and deep learning methods', *International Review of Financial Analysis*, vol. 92, article 103055, <<https://doi.org/10.1016/j.irfa.2023.103055>>.
- Breiman, L 2001, 'Random forests', *Machine Learning*, vol. 45, no. 1, pp. 5–32, <<https://doi.org/10.1023/A:1010933404324>>.
- GeeksforGeeks 2025, 'What is Exploratory Data Analysis?', GeeksforGeeks, viewed 15 March 2026, <<https://www.geeksforgeeks.org/data-analysis/what-is-exploratory-data-analysis/>>.
- Goodwin, M n.d., 'What is an API (application programming interface)?', IBM Think, viewed 15 March 2026, <<https://www.ibm.com/think/topics/api>>.
- Hochreiter, S & Schmidhuber, J 1997, 'Long short-term memory', *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, <<https://doi.org/10.1162/neco.1997.9.8.1735>>.
- Hyndman, RJ & Koehler, AB 2006, 'Another look at measures of forecast accuracy', *International Journal of Forecasting*, vol. 22, no. 4, pp. 679–688, <<https://doi.org/10.1016/j.ijforecast.2006.03.001>>.
- McNally, S, Roche, J & Caton, S 2018, 'Predicting the price of Bitcoin using machine learning', in 2018 26th Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP), IEEE, pp. 339–343, <<https://doi.org/10.1109/PDP2018.2018.00060>>.
- Murray, K, Rossi, A, Carraro, D & Visentin, A 2023, 'On forecasting cryptocurrency prices: A comparison of machine learning, deep learning, and ensembles', *Forecasting*, vol. 5, no. 1, pp. 196–209, <<https://doi.org/10.3390/forecast5010010>>.
- Sebastião, H & Godinho, P 2021, 'Forecasting and trading cryptocurrencies with machine learning under changing market conditions', *Financial Innovation*, vol. 7, no. 1, article 3, <<https://doi.org/10.1186/s40854-020-00217-x>>.
- University of Technology Sydney 2026, 41004 Analytics Capstone Project Assignment 1: plan and proposal, Canvas, viewed 15 March 2026, <<https://canvas.uts.edu.au/courses/38592/assignments/253852>>.
- Wirth, R & Hipp, J 2000, 'CRISP-DM: Towards a standard process model for data mining', in *Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining*, Manchester, pp. 29–40.
- Zhang, J, Cai, K & Wen, J 2024, 'A survey of deep learning applications in cryptocurrency', *iScience*, vol. 27, no. 1, article 108509, <<https://doi.org/10.1016/j.isci.2023.108509>>.